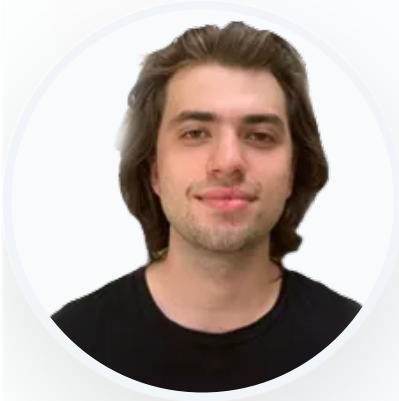




MOHAMED AL MASRI

Industrial Engineer

Industrial Engineer with experience in manufacturing process optimization, continuous improvement, and quality systems. Skilled in lean manufacturing, Six Sigma, data analysis, and CAD/CAM — bridging engineering design with production efficiency.

TECHNICAL EXPERTISE

CAD/CAM SOFTWARE

SOLIDWORKS SIEMENS NX FUSION 360 MASTERCAM
AUTOCAD PROCESS SIMULATE MOLDFLOW

QUALITY & MANUFACTURING

SIX SIGMA GREEN BELT FMEA DOE SPC GD&T
PPAP ROOT CAUSE ANALYSIS 5S LINE BALANCING
TIME STUDIES

ENGINEERING ANALYSIS

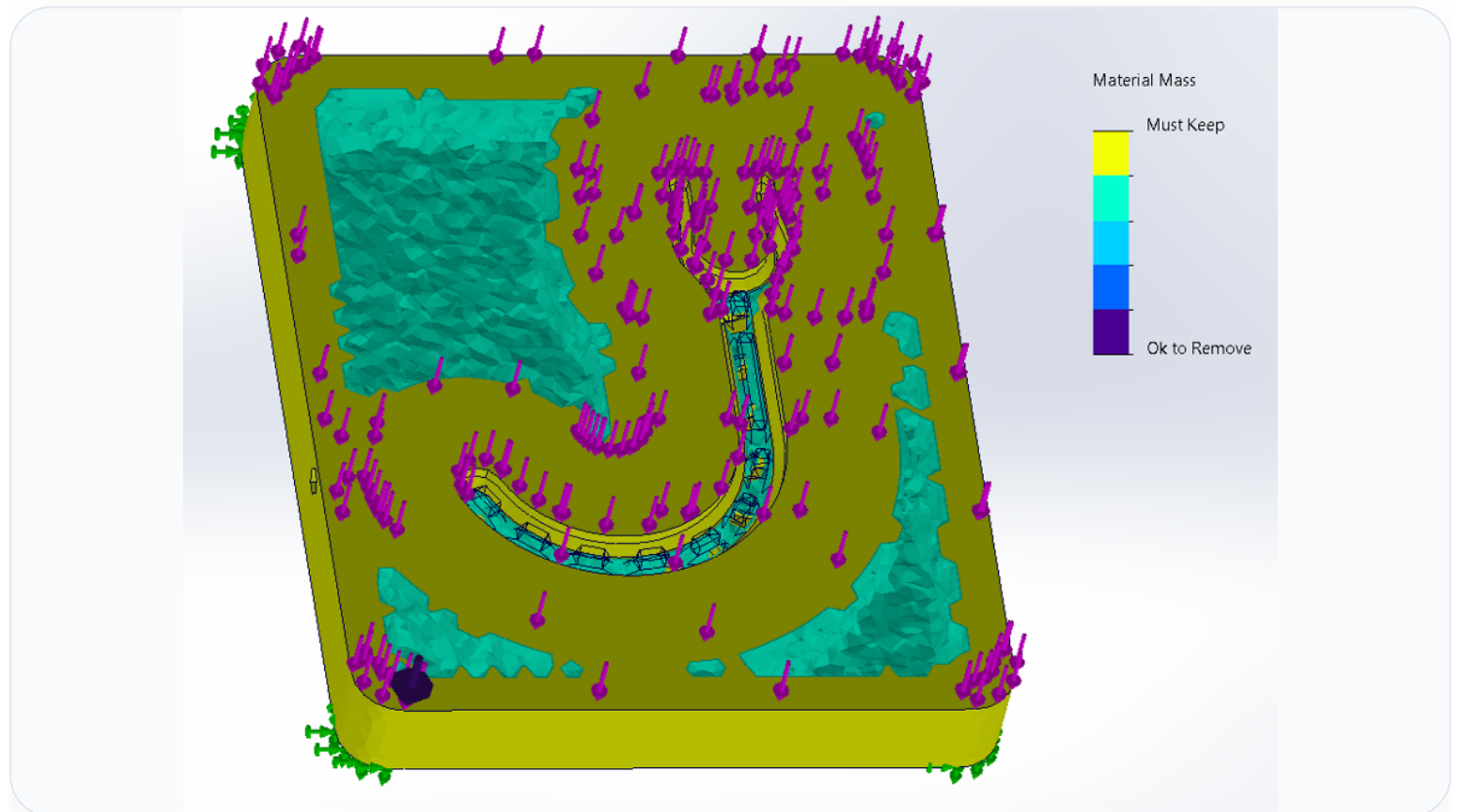
TOLERANCE ANALYSIS REVERSE ENGINEERING FEA
TOPOLOGY OPTIMIZATION PROCESS CAPABILITY ANALYSIS
ACCEPTANCE SAMPLING BOTTLENECK ANALYSIS

BUSINESS & PRODUCTIVITY TOOLS

POWER BI MS OFFICE DATA ANALYSIS
COST ANALYSIS ROI EVALUATION
DASHBOARD DEVELOPMENT

ONLINE: lowinertia.com/portfolio/user-eaa4c85c/1978/ded-manufactured_injection_mold_tooling_-_capstone__camufacturing_solutions_inc

DED-Manufactured Injection Mold Tooling - Capstone (CAMufacturing Solutions Inc.)



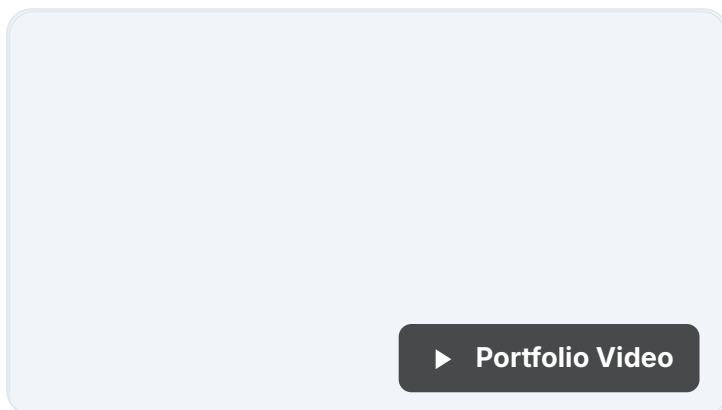
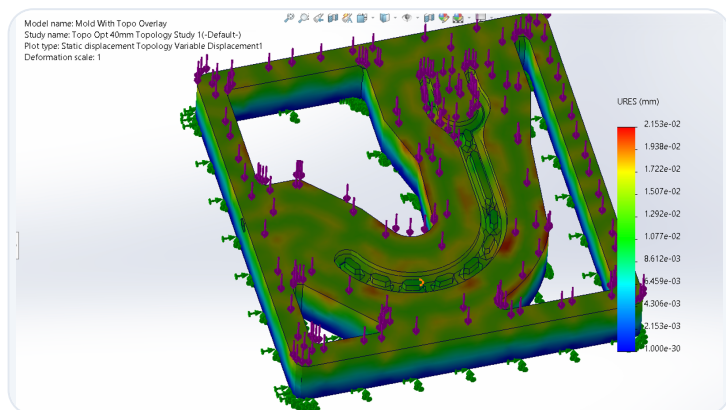
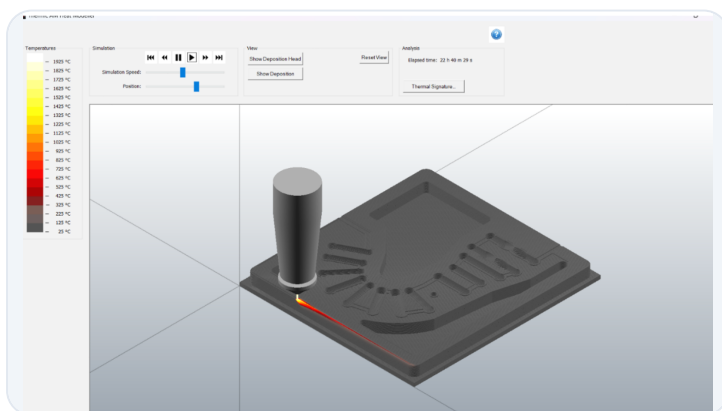
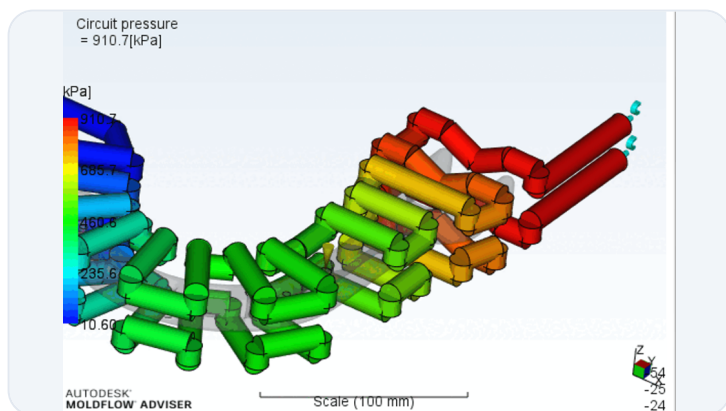
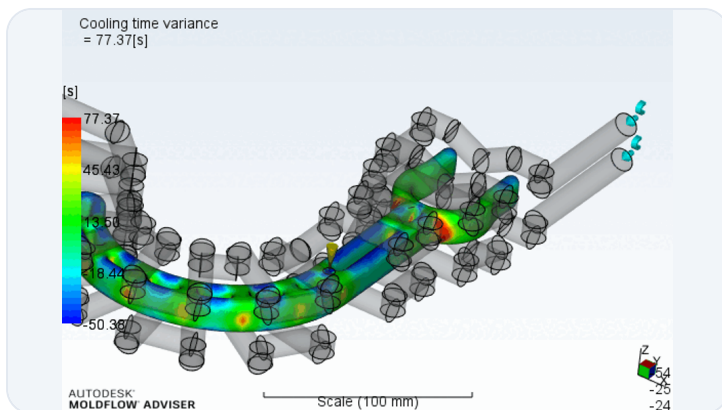
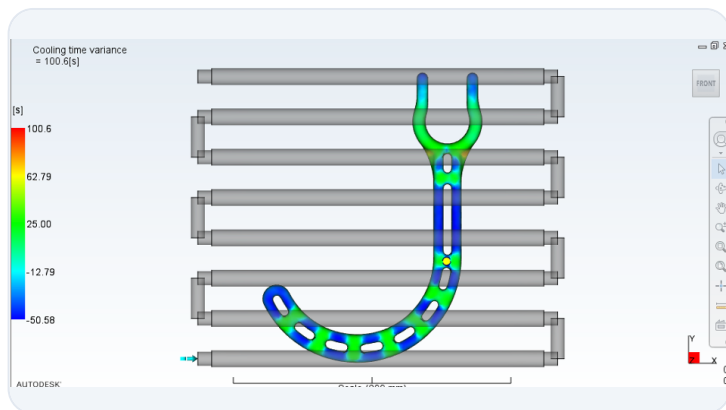
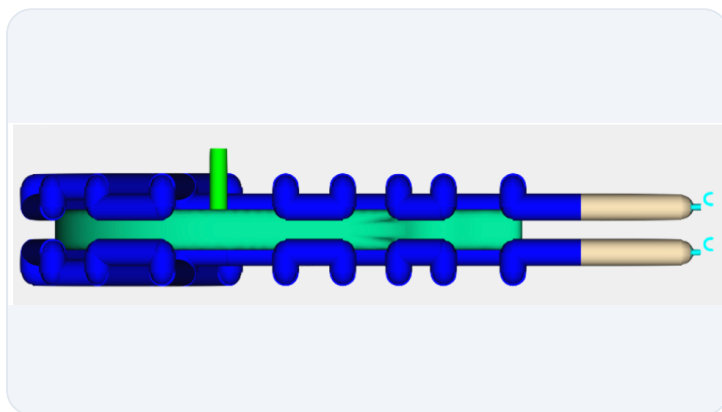
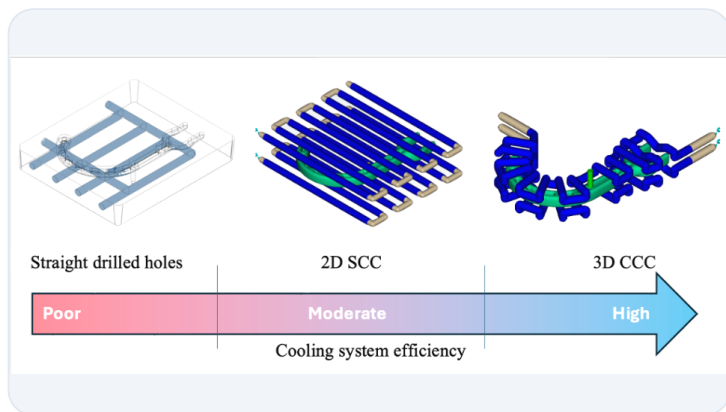
OVERVIEW

For my capstone project, I led the design of a plastic injection mold with conformal cooling using Directed Energy Deposition. Unlike traditional straight channels, this design follows part geometry, improving cooling uniformity, reducing cycle time, and minimizing warpage and defects.

★ ACCOMPLISHMENTS

- Designed a complete conformal cooling injection mold in SolidWorks for hybrid DED manufacturing on the Phillips Hybrid machine using 316L stainless steel
- Performed advanced cooling simulations in Moldflow and topology optimization, achieving **32% reduction in cycle time** and **29% lower volumetric shrinkage** compared to conventional tooling
- Conducted FEA (load, buckling, and displacement analysis) and successfully removed **51% of material** through topology optimization while maintaining structural integrity under 2 MN clamping forces
- Developed and validated complex DED toolpaths in Mastercam + APlus, implementing teardrop channel profiles and angled deposition strategies to minimize voids, manage overhangs, and ensure geometric accuracy
- Reduced projected mold build time by approximately 10 hours through optimized deposition strategies and topology improvements
- Built a full cost model and breakeven analysis demonstrating strong economic benefits driven by major cycle time savings

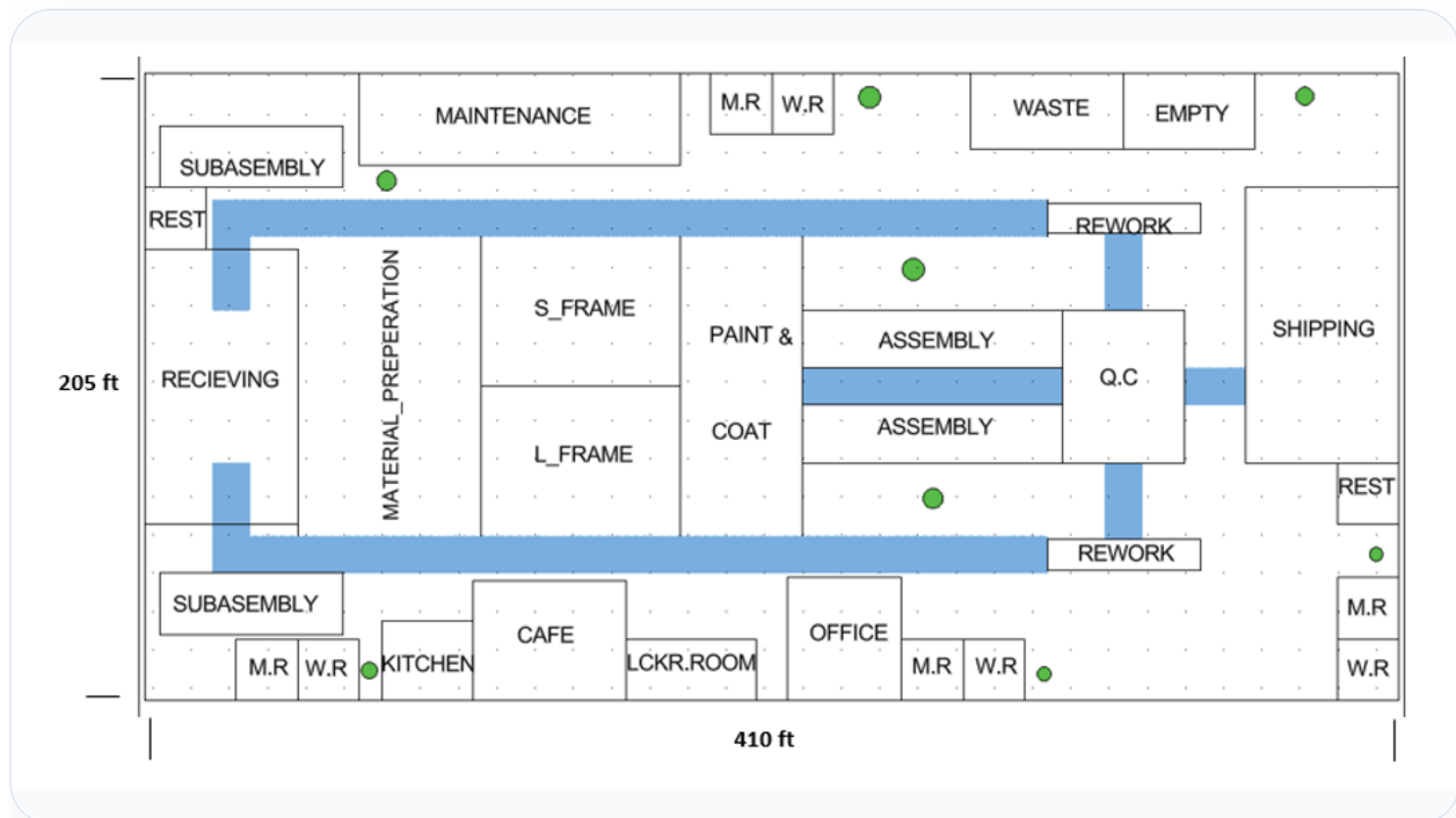
PROJECT GALLERY



► Portfolio Video

ONLINE: lowinertia.com/portfolio/user-eaa4c85c/1995/m_h_j_inc_facilities_design__manufacturing_optimization

M.H.J. Inc. Facilities Design & Manufacturing Optimization

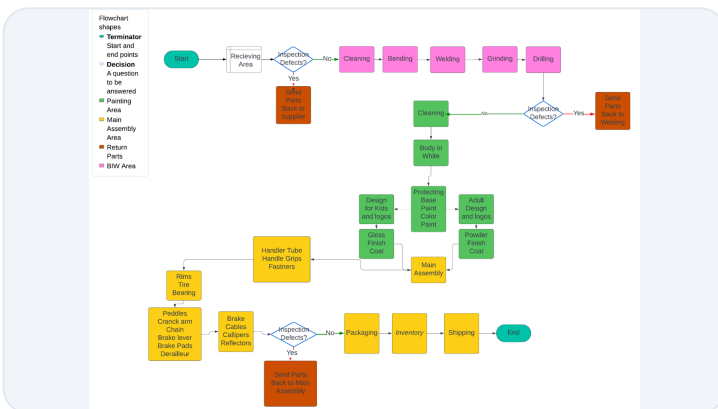
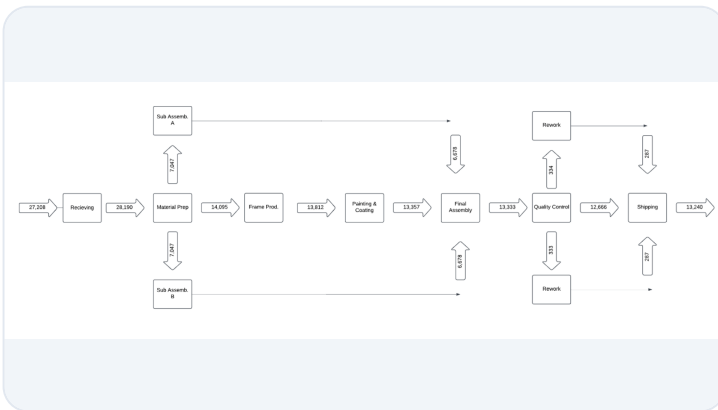


OVERVIEW

I led the design of an 85,050 sq ft bicycle manufacturing facility (M.H.J. Inc.) to produce 129,000 units annually. The project focused on creating a lean, cost-effective, and sustainable plant capable of handling seasonal North American demand with vertical integration.

- Designed the full facility layout in AutoCAD, including optimized material flow from receiving through welding, painting, assembly, rework, and shipping areas
- Conducted comprehensive production planning including takt time calculation (38.11 sec/unit), detailed line balancing, from-to charts, route sheets, and aggregate workforce planning supporting up to 130 employees
- Developed material handling strategy using forklifts and hand trucks, along with inventory and shipping logistics (8 trucks per week) to support Just-In-Time (JIT) operations
- Performed full financial analysis and cost modeling, delivering a projected **70.8% ROI** on a \$29.3M investment with annual profit of approximately \$19.5 million
- Incorporated sustainability measures including aluminum frame production, defect/rework processes, scrap handling, and recycling systems to minimize waste and environmental impact

 PROJECT GALLERY



ONLINE: lowinertia.com/portfolio/user-aaa4c85c/1980/vehicle_dashboard_assembly_simulation___work_cell_design

Vehicle Dashboard Assembly Simulation – Work Cell Design

▶ Portfolio Video

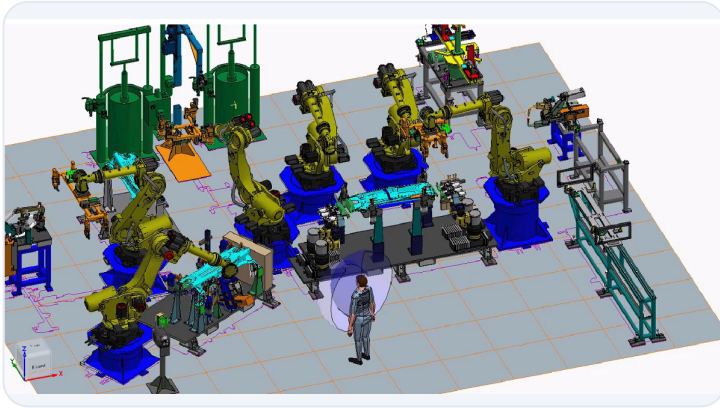
OVERVIEW

I designed and optimized a FANUC robotic work cell for high-volume automotive dashboard assembly. The system enabled flexible production for multiple vehicle variants, reducing cycle time, eliminating bottlenecks, and allowing rapid changeovers with minimal hardware changes.

★ ACCOMPLISHMENTS

- Developed a comprehensive virtual model of the full dashboard assembly production line (Work Cells 1 and 2) using Siemens Process Simulate
- Led the reconfiguration of robotic workstations (STA05, STA10, STA30) and staging areas, significantly shortening material flow distances and reducing delays
- Performed detailed robot path validation, line balancing, and bottleneck analysis to optimize cycle time and handoffs across welding, stud, and sealing operations
- Designed a flexible layout that allows Cell 01 to accommodate new product variants with minimal hardware changes
- Achieved a target throughput of **720 jobs per hour** (approximately 816,200 parts per year) while supporting multi-style vehicle production

🖼️ PROJECT GALLERY



ONLINE: lowinertia.com/portfolio/user-eaa4c85c/1981/vehicle_rear_axle_bearing_puller_design_-_apex_auto_supply

Vehicle Rear Axle Bearing Puller Design - Apex Auto Supply



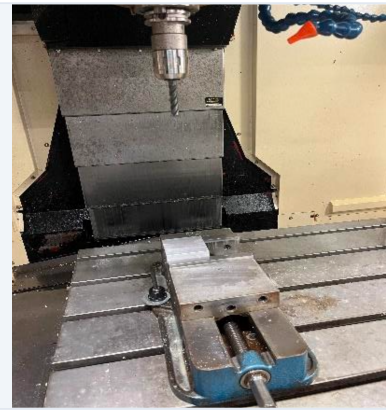
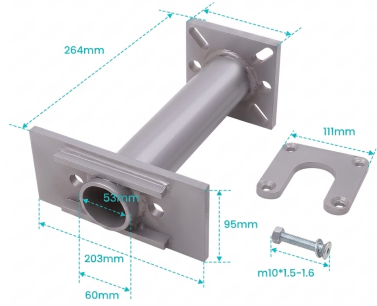
OVERVIEW

I developed a specialized rear axle bearing puller for an automotive supplier targeting Toyota 4Runner, Tundra, and Tacoma models. The tool enables safe, efficient removal of stuck bearings, reducing extraction time and risk while ensuring durability.

★ ACCOMPLISHMENTS

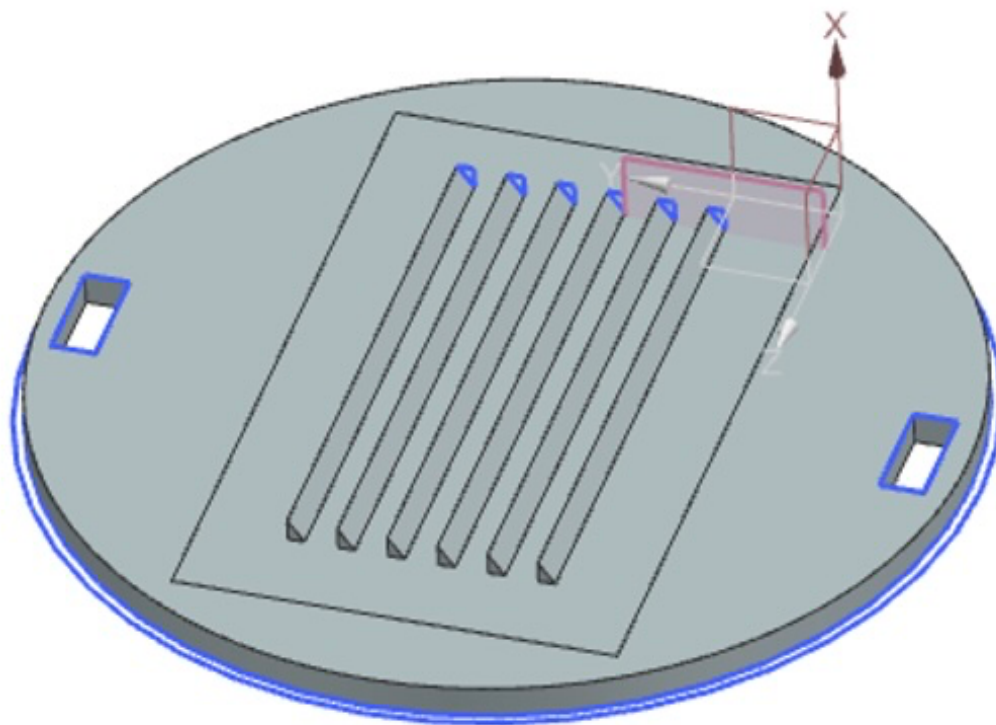
- Designed a complete rear axle bearing puller in Siemens NX using OEM specifications, flange bolt patterns, and bearing dimensions
- Engineered the tool to reliably withstand 10–20 ton hydraulic press loads using CNC-machined steel components, DOM tubing, and MIG-welded structural assemblies
- Applied strict dimensional tolerancing and GD&T on all critical mating features to ensure precise axle alignment and damage-free bearing removal/installation
- Delivered a practical, high-strength solution optimized for both performance and manufacturability that significantly simplifies a traditionally difficult and unsafe job for mechanics

PROJECT GALLERY



ONLINE: lowinertia.com/portfolio/user-eaa4c85c/1979/dfm_zebrafish_embryo_microinjection_holder

DFM Zebrafish Embryo Microinjection Holder



OVERVIEW

I redesigned a multi-component fixture for zebrafish embryo microinjection, improving alignment, assembly, and handling. The new design ensured consistent fit with standard 8.5 cm petri dishes, streamlining workflow and reducing errors in this delicate procedure.

★ ACCOMPLISHMENTS

- Redesigned a complete multi-component Zebrafish Embryo Microinjection Holder in SolidWorks using Design for Manufacturability (DFM) principles
- Incorporated self-aligning locators for rapid and accurate assembly, ergonomic handles to reduce operator fatigue, and precision toleranced features optimized for standard 8.5 cm petri dishes
- Engineered the design specifically for high-volume 3D printing, enabling cost-effective batch production while maintaining tight tolerances and excellent surface finish
- Significantly improved assembly/disassembly speed, stability, embryo accessibility, and overall usability through iterative design and prototyping
- Delivered a reliable, user-friendly fixture that streamlines laboratory workflow and reduces handling errors in precision biological research

🖼️ PROJECT GALLERY

